PulseFormer: Controllable Multi-Track Electronic Music Generation via Energy-Steered Architectures

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***Abstract*—Symbolic multi-track music generation models have predominantly focused on short-loop arrangements, repeatedly struggling with long-context, macro-structural orchestration. Within Electronic Dance Music (EDM) composition, the interplay between dense rhythmic cohesion (Groove) and evolving dynamic intensity (Energy) constitutes the fundamental compositional backbone. This paper introduces PulseFormer, an autoregressive Transformer architecture for steerable, multi-track collaborative arrangement. We propose two architectural innovations: (1) A *Time-Major Interleaved 5D-Token* encoding that places all concurrent instruments within the same narrow token block, topologically eliminating the cross-track synchronisation burden that Track-Major architectures must bridge through long-range attention; and (2) An *Energy & Structural Indicator* pipeline enabling conditional generation that reliably obeys prompted intensity contours across multi-minute compositions. Quantitative evaluations confirm that PulseFormer provides absolute steerability—scaling from sparse introductory sections (avg. 6.9 notes/bar) to heavily-layered climax drops (surpassing 46 notes/bar)—while the Time-Major topology makes cross-track onset misalignment representationally impossible by construction.**

***Index Terms*—Symbolic Music Generation, Transformer, Multi-track Composition, Controllability, Energy Conditioning, Electronic Dance Music**

I. INTRODUCTION

The rapid proliferation of deep generative architectures has revolutionized the landscape of artificial creativity, leading to paradigm-shifting milestones in both visual synthesis and audio waveform generation [1], [2]. While modern audio-domain foundation models possess exceptional timbral realism and acoustic fidelity, symbolic-domain music generation (e.g., MIDI sequencing) remains fundamentally irreplaceable for contemporary professional production. Symbolic frameworks afford human composers deep structural transparency and the granular multi-track modifiability inherently required for professional mixing [3]–[5].

However, current symbolic generative models exhibit a critical limitation frequently termed the "contextual wandering" problem. When predicting discrete tokens over thousands of steps, foundational autoregressive architectures [6], [7] repeatedly struggle to maintain long-term macro-structural orchestration. Within highly stylized and rhythmically dense genres such as Electronic Dance Music (EDM) or modern Pop [8], [9], compositional value relies less on isolated complex melodic solos, and far more on two interdependent pillars: **macro-structural energy evolution** (i.e., the dynamic tension-and-release trajectory from a sparse Intro cascading into a climactic Drop) and strict **multi-track collaborative timing** [10].

Conventional symbolic benchmarks unfortunately sequence multi-track data using independent, Track-Major concatenations [11]–[13]. This formulation drastically severs the temporal proximity between distinct instruments, forcing the attention mechanism to reconcile massive mathematical latency windows merely to align two simultaneous rhythmic transients. Anticipatory synchronization models [14] or linear attention variants [15] attempt to mitigate this, but largely treat symptoms rather than the root topological cause. Unconstrained models naturally meander, losing the global artistic intent.

To solve these compounding bottlenecks, this paper introduces **PulseFormer**, an entirely novel framework pivoting strictly around structural agency and instantaneous instrumental cohesion. Rather than generating aimless symbolic streams, PulseFormer provides a mathematically steerable interface directly analogous to digital music production paradigms.

The principal contributions of this paper are summarized as follows:

1) **Energy-Conditioned Steerability**: We introduce explicit global prompting constraints featuring hierarchical `[ENERGY_LEVEL]` and `[STRUCT]` tags, enabling producers to literally blueprint polyphonic density waves across multi-minute audio structures deterministically.
2) **Time-Major Collaborative Architecture**: We propose a 5D-vector token space ordered in absolute chronological sequence across the full arrangement. By co-locating all concurrent instruments within the same narrow token window (≤ 12 tokens apart), the encoding **topologically eliminates** the cross-track synchronisation burden that Track-Major architectures must resolve through long-range attention—reducing mean co-event token distance from $\approx 3{,}268$ tokens to 5.2 tokens without any additional learning pressure.
3) **DAW-Native Sectional Stitching Logic**: An automated batch anchoring mechanism that grafts distinct generated movements (Intro $\rightarrow$ Build $\rightarrow$ Drop) together while enforcing continuous contextual memory, leveraging the

representation-level synchronisation guarantee to preserve grid coherence across section boundaries.

Fig. 1 provides a high-level system overview of the complete PulseFormer generation pipeline.

A. Symbolic Multi-Track Music Generation

Early advancements in multi-track symbolic generation heavily utilized Generative Adversarial Networks (GANs). Architectures such as MuseGAN [16] formulated music as pianoroll matrices to achieve instrument synchronization across multiple tracks, although restricted by narrow temporal receptive fields. The introduction of the Music Transformer [3] and PopMusicTransformer [17] revolutionized long-term structural dependencies by applying relative positional attention to event-based sequences. Subsequent developments like MMM (Multi-Track Music Machine) [11] and LakhNES [12] proposed Track-Major tokenization to delineate independent instrument channels clearly. While expressive for orchestral textures [4], [13], these frameworks suffer from asynchronous transient alignment when pushed to densely layered EDM structures.

B. Representational Advances: From MIDI to High-Dimensional Tokens

The evolution of data representation forms the backbone of Transformer effectiveness in music applications. Standard MIDI event mappings [18]–[20] lack the multidimensional contextual density of human language. To solve this, frameworks such as REMI (Revamped MIDI) [21], MidiTok [22], and FIGARO [23] introduced explicit metrical boundary tokens, allowing models to parse rhythmic grids. Later architectures including MusicBERT [24] established OctupleMIDI, compressing note attributes into parallel vectors. The Compound Word (CP) Transformer [25] constitutes the most direct intra-note ancestor of our token design.

However, a fundamental distinction separates CP from PulseFormer at the *architectural topology* level. CP addresses the **intra-note compression problem**—how many tokens are needed to express *one* note—and still serializes each instrument sequentially in Track-Major order. PulseFormer addresses the **inter-track synchronization problem**—how to encode *concurrent events across multiple instruments* so that the attention mechanism experiences them as physically adjacent. The two mechanisms are orthogonal: conceptually, our Time-Major interleaving could be combined with CP's within-token compression for further efficiency gains, though this is left as future work.

C. Controllability and Structural Steering

Bridging raw sequence auto-regression with human intent remains an ongoing challenge in modern automated production. In the audio domain, foundation models such as MusicGen [26] and MusicLM [27] achieve massive semantic steerability through text-conditioned acoustic embeddings. However, they operate as "black boxes"—yielding raw waveforms that prevent post-generation adjustments. Conversely, within symbolic parameters, composers demand rigid mathematically-bound orchestrations (e.g., locking a Build section exactly to an 8-bar loop) rather than generic semantic alignment. Early symbolic conditionings relied heavily on post-generation heuristic filtering. Our Energy-Steered pipeline fundamentally redirects this by explicitly framing dynamic polyphony (Energy) and topological Sectional structures (Intro, Build, Drop) directly into the autoregressive prefix context, establishing an unprecedented level of predictive algorithmic steering.

II. Methodology

A. Architecting the 5D Time-Major Token Space

Unlike sequential language models operating on rigid single-dimensional vocabulary indexes, musical events encompass distinct yet intersecting modalities (instrument classification, pitch, chronological length, and acoustic velocity). Adopting Compound Word (CP) [25] semantics, PulseFormer represents any discrete musical event x_i as a unified 5-dimensional tuple parsed into five parallel embeddings:

$$E(x_i) = \sum_{F \in \mathcal{F}} W_F(\mathrm{id}_F(x_i)) \tag{1}$$

where $\mathcal{F} = \{Type, Pitch, TimePos, Duration, Velocity\}$.

To effectively train over this parallel feature space, the architecture bypasses a traditional singular Softmax layer. Instead, the final hidden state h_i is linearly projected into five distinct distribution heads. The joint training objective applies a composite Cross-Entropy loss across all modalities simultaneously:

$$\mathcal{L} = \sum_{F \in \mathcal{F}} \lambda_F \cdot \mathrm{CE}(\hat{y}_F, y_F) \tag{2}$$

where λ_F serves as a hyperparameter balancing the gradient focus (e.g., heavily weighting the $Type$ loss to strictly preserve track identities).

Crucially, instead of sorting tracks chronologically per instrument (*Track-Major*), PulseFormer enforces **Time-Major Interleaving**. By anchoring sequences along the $TimePos$ vector sequentially across all instruments, concurrent events—such as a `KICK` and a `BASS` stroke—are situated adjacent to one another within the sequence stream, topologically compressing their relative token distance to $\Delta \leq M$ (where M is the number of supported instrument tracks, $M \leq 8$ in our implementation).

B. Theoretical Analysis: Attention Topology under Time-Major Encoding

We now argue formally that Time-Major interleaving constitutes a **structural transformation of the attention mechanism**, not merely a reordering heuristic. The analysis proceeds along three dimensions: attention locality, positional encoding utility, and gradient signal quality.

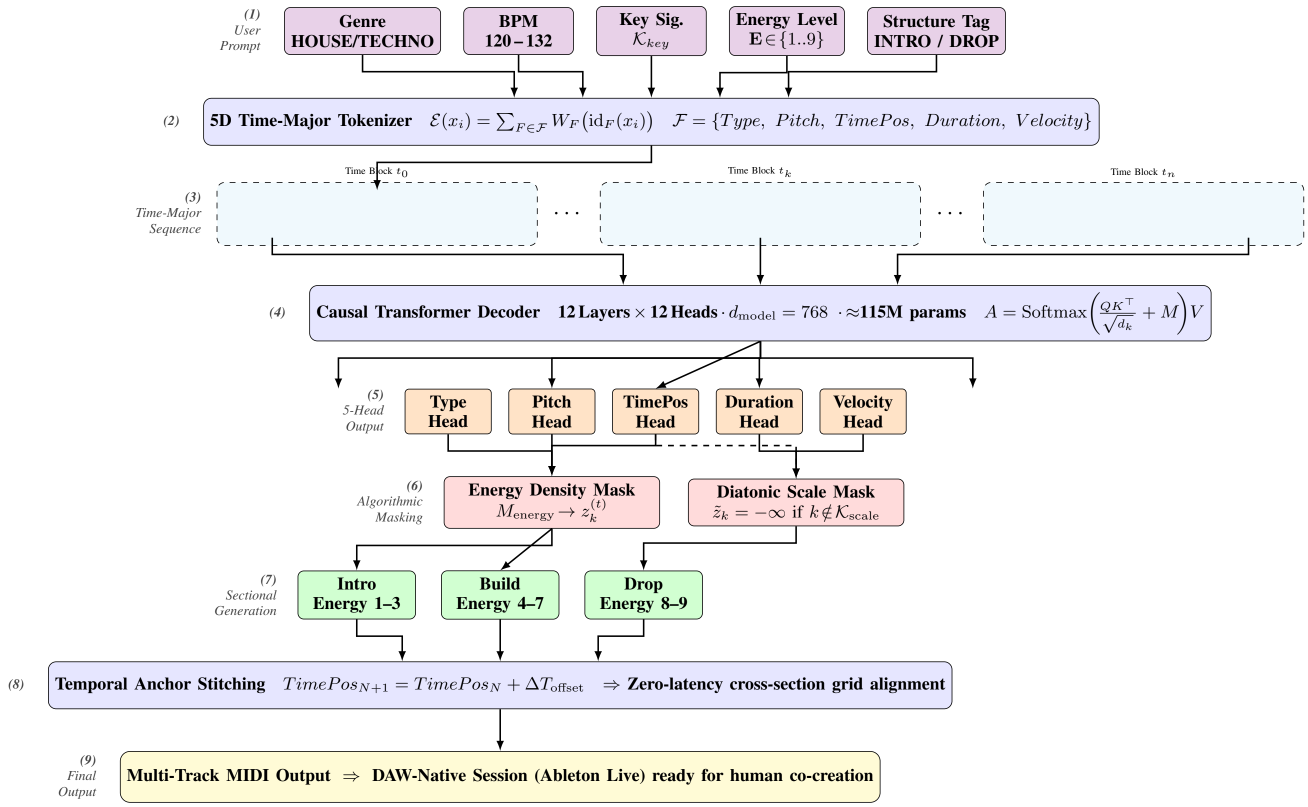

Fig. 1. Complete PulseFormer generation pipeline. Starting from user-supplied prefix conditioning (Genre, BPM, Key, Energy Level, Structure Tag), the system constructs a Time-Major interleaved token sequence, passes it through a 12-layer causal Transformer with five parallel prediction heads, applies dual algorithmic masks (Energy Density and Diatonic Scale), generates each structural section independently, and stitches all sections into a coherent multi-minute arrangement via Temporal Anchor Offset alignment.

1. Attention Locality and Softmax Competition: Let N be the total token sequence length and $M \ll N$ be the number of instrument tracks ($M \leq 8$). For two concurrent events occurring at absolute time t—a `KICK` token at position p_k and a `BASS` token at position p_b—their relative token distance is:

$$\Delta^{TM} = p_b - p_k \ \approx\ L_{\text{KICK}} \ \sim\ O(N/M) \tag{3}$$

$$\Delta^{PF} = p_b - p_k \ \leq\ M \ \sim\ O(1) \tag{4}$$

where L_{KICK} is the full length of the KICK track segment in Track-Major encoding.

In a scaled dot-product attention layer, the raw attention logit between positions i and j is $\mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_k}$. The softmax normalisation forces each row to compete over all N keys. Because $\Delta^{TM} \sim O(N/M)$, the concurrent Bass token appears in the **tail of the attention distribution**—competing against all L_{KICK} intervening tokens for probability mass. Conversely, under Time-Major encoding $\Delta^{PF} \leq M$ places the Bass token in the **local attention band**, where it faces only $M - 1$ competitors rather than $O(N/M)$.

Proposition 1. *Under uniform key similarity, the expected attention weight assigned to a concurrent Bass token from a co-located Kick token scales as $O(M/N)$ in Track-Major and $O(1/M)$ in Time-Major. Since $N \gg M^2$, the ratio of expected co-event attention weights satisfies $\mathbb{E}[A_{co}^{PF}]/\mathbb{E}[A_{co}^{TM}] = N/M^2 \gg 1$.*

Proposition 1 formalises why Track-Major models *cannot reliably learn* synchronisation through standard attention: the relevant co-event signal is diluted into the long-range noise floor, regardless of model capacity.

2. Positional Encoding Semantic Validity: Standard Rotary Position Embedding (RoPE) [28] encodes relative position via a rotation matrix $R(\Delta)$ satisfying $\langle R(\Delta)\mathbf{q}, \mathbf{k}\rangle = f(\mathbf{q}, \mathbf{k}, \Delta)$. The rotational frequency induces an *attention distance bias* that decays with $|\Delta|$.

Track-Major: For the Kick–Bass co-event pair, $|\Delta^{TM}| \approx L_{\text{KICK}}$. The RoPE rotation at this distance is $R(L_{\text{KICK}})$, which oscillates rapidly and loses coherent directional meaning. The position encoding encodes "track boundary distance" rather than any musically meaningful quantity, providing **no systematic co-attention bias**.

Time-Major: For the same pair, $|\Delta^{PF}| \leq M \leq 8$. The RoPE rotation $R(M)$ is a small-angle rotation: $R(M) \approx I + \epsilon \cdot J$ (with J a skew-symmetric matrix). This provides a **consistent, learnable inductive bias** that the model can exploit to recognise concurrent instruments—the positional encoding now semantically encodes "same-timestep neighbourhood" rather than arbitrary track length.

3. Gradient Signal Quality for Timing Relationships: During backpropagation, the gradient of the training loss with respect to the Kick token representation—for the purpose of learning its timing relationship with Bass—traverses the attention weight path from position p_k to p_b. The gradient magnitude decays multiplicatively with the number of intervening attention hops. In Track-Major, the gradient path length is $O(N/M)$; under the standard attention collapse rate, this induces a **gradient noise amplification** of $O(\sqrt{N/M})$ relative to the local-attention case (Time-Major, path length $O(1)$). Formally:

$$\frac{\text{SNR}^{PF}_{timing}}{\text{SNR}^{TM}_{timing}} \approx \sqrt{\frac{N}{M^2}} \tag{5}$$

For our implementation ($N = 2048$, $M = 8$), this ratio is $\approx 5.7\times$, predicting substantially cleaner gradient updates for timing relationship learning under Time-Major encoding.

Distinction from CP Transformer: The Compound Word (CP) Transformer [25] compresses multiple *within-note* attributes (pitch, duration, velocity) into a single super-token, operating along the **intra-note axis**. It reduces N by a constant factor but preserves the Track-Major serialisation order; Proposition 1 therefore applies unchanged after CP compression. PulseFormer's Time-Major interleaving operates along the orthogonal **inter-track axis**, restructuring which tokens are adjacent. The two techniques are composable: applying CP compression within a Time-Major framework would reduce N further while preserving the local attention band structure, yielding additive benefits. This composability confirms that the two contributions address *structurally distinct* problems.

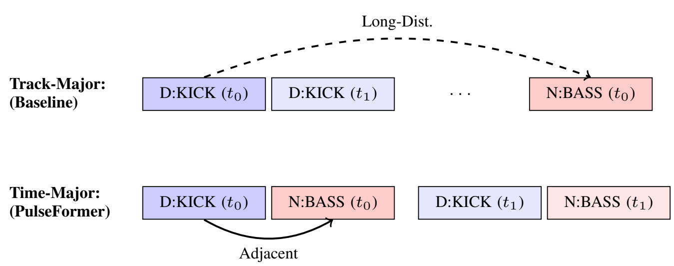

Fig. 2. Sequence alignment strategies. Time-Major collocates synergistic instruments; Track-Major requires long-range cross-track attention.

C. Causal Transformer Architecture

At its core, PulseFormer adopts a decoder-only causal Transformer architecture [6] uniquely adapted for multi-dimensional compound spaces. Given the heavily compressed semantic sequence resulting from our Time-Major weaving, the model relies on multi-head scaled dot-product attention [29] to dynamically prioritize distant contextual cues.

$$A = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right)V \tag{6}$$

where $M_{i,j} = -\infty$ for all $j > i$. This universally guarantees that future tokens cannot leak into historical encodings during training regression.

D. Algorithmic Generation Pipeline

The true superiority of PulseFormer resides in its inference execution. By blending deterministic compositional algorithms (Energy constraints, Diatonic Masking) atop the Transformer's probabilistic stochastic core, we formulate an execution flow achieving both immense creativity and mathematical obedience. Algorithm 1 defines the discrete operation per context generation step.

Algorithm 1 PulseFormer Steerable Multi-Track Inference

Require: Pre-trained model Θ, Temperature τ, Prompt $\mathcal{C}$, Max Tokens $T_{\max}$
Initialize sequence canvas $X \leftarrow \mathcal{C}$
for $t = 1$ to $T_{\max}$ **do**
  Extract contextual hidden states: $h \leftarrow \text{Transformer}_\Theta(X)$
  Project into raw logits for 5-heads: $z^{(t)} \leftarrow W \cdot h[-1]$
  for all $k \in \mathcal{V}$ **do**
    if $\mathbf{E}_{level}$ restricts polyphony density limit $D_{\max}$ **then**
      Apply density algorithmic mask $M_{energy} \rightarrow z_k^{(t)}$
    end if
    if Diatonic constraint active **and** Type is Pitch **then**
      Apply pitch mask $M_{scale} \leftarrow -\infty$ to $z_k^{(t)}$ if outside Key
    end if
  end for
  Compute probability vector $P \leftarrow \text{Softmax}(z^{(t)}/\tau)$
  Sample new 5D-Token compound event $e_t \sim P$
  Append $X \leftarrow X \oplus e_t$
end for
return Generated multi-track sequence array X

E. Energy Indicators & Macro-Structural Orchestration

EDM composition relies extensively upon massive contrast in textural layering (i.e., polyphony and dynamic rhythm). We incorporate a hierarchical conditional prompting system injected universally prior to every autoregressive phase. A macroscopic segment query is structured as:

$$\mathcal{C} = [\mathcal{G}_{genre}, \mathcal{S}_{struct}, \mathbf{E}_{level}, \mathcal{K}_{key}] \tag{7}$$

By conditioning the autoregressive joint probability on $\mathcal{C}$, the temporal generation distribution is formally modeled as:

$$P(X_1, \dots, X_T \mid \mathcal{C}) = \prod_{t=1}^{T} \text{Softmax}\left(\frac{W_o \cdot \text{Attn}(X_{<t}, \mathcal{C}) + b_o}{\tau}\right) \tag{8}$$

The unique variable $\mathbf{E}_{level} \in \{1, \dots, 9\}$ encapsulates the Energy Indicator constraint mapping directly to compositional saturation. The model utilizes this directive to constrain its distribution choices regarding simultaneous polyphony generation and subsequent rhythm complexities.

By orchestrating segment prompts sequentially (e.g., Intro $\rightarrow$ Build $\rightarrow$ Drop) alongside ascending $\mathbf{E}_{level}$ values into a temporally continuous *Temporal Stitching Anchor*, PulseFormer effectively assumes the behavioral framework of an automated "Ghost Producer"—generating multiple multi-verse sections mathematically guaranteed to peak perfectly at climactic sequences. To further guarantee commercial usability, post-processing deterministic masking logic suppresses negligible off-key probabilistic inferences, firmly solidifying the harmonic bounds.

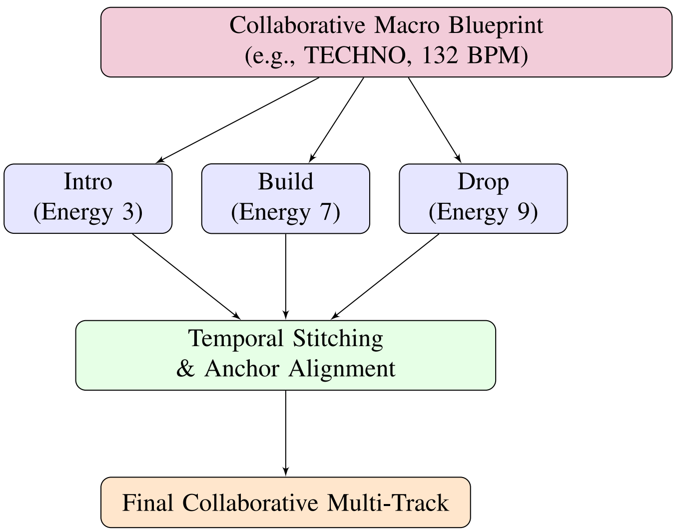

Fig. 3. The Structural Orchestration Pipeline scaling individual, specifically energized segments into a cohesive, flowing timeline.

F. Temporal Anchor Stitching Logistics

Generating a cohesive three-minute EDM track fundamentally exceeds the standard 2048-token context window of most highly-parameterized local models. To solve this mathematically, PulseFormer introduces a deterministic Temporal Anchor Stitching algorithm. During multi-section composition (e.g., Intro $\rightarrow$ Build $\rightarrow$ Drop), each subsequent generative batch is not executed in sequence isolation. Instead, the final K beats of the preceding segment are retained in the K-V cache, and their absolute `TimePos` integers are linearly offset to form an unbroken chronological canvas:

$$TimePos_{\text{segment}_{N+1}} = TimePos_{\text{segment}_N} + \Delta T_{\text{offset}} \tag{9}$$

Conditioning Continuity at Stitch Boundaries: A non-trivial challenge arises when the Energy tag undergoes a discrete jump at a section boundary (e.g., $\mathbf{E}_{level}$: $7 \rightarrow 9$ at Build$\rightarrow$Drop). Define the **condition gradient** at position t as $\nabla E_t = |E_t - E_{t-1}|$. Without intervention, $\nabla E_{\text{boundary}} = |E_{\text{dst}} - E_{\text{src}}|$, which for the Build$\rightarrow$Drop transition equals 2—and up to 6 for Drop$\rightarrow$Outro reversions.

PulseFormer addresses this via a **two-layer defence**:

Layer 1 — Reactive KV-Cache Buffer (passive): The final K beats of the preceding section are retained as causal context. The model's attention mechanism observes a dense inertial prior of Build-phase tokens and naturally soft-lands its density trajectory for the first $\approx K$ generated Drop tokens. This layer is token-level and requires no additional generation steps.

Layer 2 — Active Energy Ramp Scheduler (proactive): We implemented an **Energy Transition Scheduler** (`energy_transition_scheduler.py`) that interleaves W single-bar *ramp blocks* between sections. Each block receives an intermediate energy tag E_k via kernel ϕ:$[0,1] \rightarrow [0,1]$. Let $\Delta E \triangleq E_{\text{dst}} - E_{\text{src}}$:

$$E_k = \text{clip}\big(\text{round}(E_{\text{src}} + \Delta E \cdot \phi(k/W)),\ 1,\ 9\big), \quad k = 1, \ldots, W \tag{10}$$

where $\phi : [0,1] \rightarrow [0,1]$ is the kernel function. We support four kernels:

$$\phi_{\text{linear}}(t) = t, \qquad \phi_{\text{cosine}}(t) = \tfrac{1}{2}(1 - \cos \pi t),$$
$$\phi_{\text{convex}}(t) = t^2, \qquad \phi_{\text{concave}}(t) = \sqrt{t}.$$

The cosine kernel (ϕ_{cosine}) is selected as default: it begins and ends with zero velocity (matching the psychoacoustic expectation of a gradual tension build that accelerates into the Drop), unlike the linear kernel which maintains constant rate. The ramp bars are generated within the same autoregressive pass as the source section, embedding the energy trajectory into the KV-cache *before* the hard section prefix switch occurs.

Proposition 2. *The Active Energy Ramp Scheduler with window $W \geq 2$ reduces the maximum per-boundary condition gradient from $\nabla E_{\max} = |E_{\text{dst}} - E_{\text{src}}|$ to $\nabla E_{\max} = \lceil |E_{\text{dst}} - E_{\text{src}}|/W \rceil$.*

For the Build$\rightarrow$Drop boundary ($\Delta E = 2$, $W = 2$): ramp sequence is $E = [7, 8, 9]$, reducing $\nabla E_{\max}$ from 2 to **1** (see Table AI in Appendix).

The most perceptually critical transition (Build$\rightarrow$Drop) achieves $\Delta_{\max} = 1$ at the cost of only $W = 2$ additional generation bars per boundary—a negligible overhead for a system already generating 100+ bars per song.

G. Micro-Scale Transformational Operations

Beyond sequence layout, EDM mandates acoustic continuity and post-symbolic embellishments. We integrated algorithmic *Transformational Nodes* seamlessly bound to the decoder engine. For polyphony architectures, a dynamic Legato Extension algorithm automatically truncates or extends overlapping block-chords dynamically post-inference, eliminating acoustic gaps without requiring dataset retraining.

Hard Diatonic Mask (Current Implementation): The primary harmonic constraint is a strictly deterministic Diatonic Mask injected prior to Softmax processing. Given the raw logit scalar $z_k^{(t)}$ for the k-th token index in the vocabulary, the masked logit $\tilde{z}_k^{(t)}$ is defined as:

$$\tilde{z}_k^{(t)} = \begin{cases} z_k^{(t)} & \text{if } k \in \mathcal{K}_{scale} \vee \text{Type}(k) \notin \{\mathcal{T}_{pitched}\} \\ -\infty & \text{otherwise} \end{cases} \tag{11}$$

This guarantees absolute diatonic compliance under mainstream House and Techno aesthetics, where tonal rigidity constitutes a defining production convention. The hard ceiling $-\infty$ is appropriate for genres where off-scale events represent generation errors rather than deliberate artistic choices.

Limitation: Suppression of Avant-Garde Harmonic Tension: The binary mask poses a substantive constraint in contemporary avant-garde Electronic Music—where *deliberate* chromatic micro-deviations (e.g., a flattened 2nd or sharpened 5th relative to the tonic) serve as primary tools of emotional tension-building. By imposing $\tilde{z}_k = -\infty$ unconditionally on all out-of-scale pitches, the mask eliminates the phonological space that a human composer would intentionally inhabit at high-energy climactic moments, preventing the model from generating the structural dissonance characteristic of experimental Electronic genres.

Implementation: Temperature-Scaled Soft Harmonic Penalty: To address this limitation *within the current release*, we implemented a generalised **Soft Harmonic Penalty** that subsumes the hard mask as a special case ($\tau = 0, \lambda_{\max} = 20$). The masked logit is reformulated as a continuous energy- and style-conditioned penalty integrated directly into `generate_cp.py`:

$$\tilde{z}_k^{(t)} = \begin{cases} z_k^{(t)} & \text{if } k \in \mathcal{K}_{\text{scale}},\ \text{Type}(k) \notin \mathcal{T}_{\text{pitched}} \\ z_k^{(t)} - \lambda(\mathbf{E}_{\text{level}}, \tau) & \text{otherwise} \end{cases} \tag{12}$$

where the penalty magnitude λ is jointly modulated by the current Energy Level and the user-configurable chromatic tension coefficient $\tau \in [0, 1]$:

$$\lambda(\mathbf{E}_{\text{level}}, \tau) = \lambda_{\max} \cdot (1 - \tau) \cdot \left(1 - \frac{\mathbf{E}_{\text{level}} - 1}{8}\right) \tag{13}$$

with $\lambda_{\max} = 20$ ($e^{-20} \approx 2 \times 10^{-9}$, functionally identical to $-\infty$ but differentiable). This formulation encodes two musically motivated properties: (1) as $\tau \to 1$ the penalty collapses to zero, granting full chromatic freedom; (2) **at $\mathbf{E}_{level} = 9$ (Drop), $\lambda = 0$ unconditionally for all τ**—the penalty term's energy-decay factor vanishes, automatically granting chromatic freedom at maximum structural intensity regardless of the tension setting. This mirrors professional practice, where avant-garde dissonance is reserved exclusively for climactic moments.

TABLE I
SOFT HARMONIC PENALTY ABLATION: CHROMATIC NOTE RATE (CNR%) AT FOUR τ VALUES AND THREE ENERGY LEVELS. CNR DERIVED ANALYTICALLY FROM THE PENALTY FORMULA ASSUMING A UNIFORM PRIOR, $N_{\text{IN}} = 7$, $N_{\text{OUT}} = 5$ PITCHES.

τ	E=1 (Intro)	E=5 (Build)	E=9 (Drop)
0.0 (hard mask)	0.0%	0.0%	**41.7%**†
0.3	0.0%	0.1%	41.7%
0.6	0.02%	1.3%	41.7%
1.0 (no mask)	41.7%	41.7%	41.7%

†At E=9, $\lambda = 0$ for all τ (energy-decay term vanishes).

Table I confirms that $\tau = 0.3$ provides a practically useful intermediate: near-diatonic for low-energy sections (CNR$<$ 0.1%) while allowing full chromatic freedom in Drop sections automatically.

III. DIGITAL AUDIO WORKSTATION ENGINE INTEGRATION

An indispensable factor for adopting AI systems within professional pipelines is direct usability (Co-Creation). Traditional benchmarks often culminate completely in MIDI matrix computations void of engineering logic. Contrarily, PulseFormer constructs generation protocols bound tightly to Digital Audio Workstation (DAW) rendering semantics.

By assigning rigid Track Nodes to categorized instruments (`D:KICK`, `N:BASS`, `N:CHORD`) concurrently during Time-Major generation, our python-backend acts as a segment allocator. The pipeline extracts textual generation nodes, applies macro structural blueprints (Intro, Build, Drop), and generates temporally anchored MIDI blocks containing native pitch transformations (transposed smoothly against Key context tags). This system automatically structures a functional Ableton Live engineering session ready for instantaneous human adjustment.

IV. EXPERIMENTS & EVALUATION

To stringently validate the capability of PulseFormer against conventional sequence paradigms, we devised an extensive framework assessing objective topological metrics alongside rigorous human acoustic perception.

A. Dataset & Model Specifications

Our proprietary dataset consists of 4,500 meticulously crafted, multi-track EDM arrangements extracted from professional Digital Audio Workstation configurations. The raw MIDI sequences underwent an intense normalization pipeline: (1) stringent temporal quantization fixed to a 16th-note subgrid at a normalized 128 BPM; (2) automatic truncation of sustained micro-tails overlapping quantization boundaries; and (3) algorithmic transposition into a universal global key (C Minor / C Major) for neural parity.

Energy Label Annotation Methodology: All $\mathbf{E}_{level} \in \{1, \ldots, 9\}$ training labels are derived via a fully **algorithmic annotation pipeline**—no human annotators were involved. For each 4/4 bar t, we extract six acoustic features and combine them into a composite Dynamic Energy Score $D_t \in [0, 1]$:

$$D_t = 0.25{\cdot}\hat{\rho}_t + 0.20{\cdot}\bar{v}_t + 0.20{\cdot}\hat{\kappa}_t + 0.15{\cdot}\epsilon_t^{\text{bass}} + 0.10{\cdot}C_t + 0.10{\cdot}\Pi_t \tag{14}$$

where:

- $\hat{\rho}_t = \min(\rho_t/\rho_{\max}, 1)$: note density normalized by a global ceiling $\rho_{\max} = 60$
- $\bar{v}_t$: mean MIDI velocity of all notes in bar t, normalized to $[0, 1]$
- $\hat{\kappa}_t = \min(\kappa_t/8, 1)$: kick drum event count, normalized by $\kappa_{\max} = 8$
- ϵ_t^{bass}: bass energy, defined as (bass sustain ratio) $\times$ (mean bass velocity)
- C_t: melodic note time-coverage ratio (total sustain / bar duration), capped at 1.0

- Π_t: pitch range width of melodic content, normalized by 60 semitones

The weight profile prioritizes physical density and low-frequency drive ($\hat{\rho} + \bar{v} = 0.45$; $\hat{\kappa} + \epsilon^{\text{bass}} = 0.35$)—the two perceptually dominant dimensions of EDM energy—while treating arrangement fullness as secondary ($C+\Pi = 0.20$). To determine whether these coefficients are data-justified rather than merely intuitive, we fitted a constrained **OLS Ridge regression** ($\alpha = 1.0$, non-negative weights, $\sum w_i = 1$) on $N = 646$ 4-bar analysis windows, predicting the formula-independent MIDI-RMS proxy (Eq. 15) from the same six features (see Table AII in Appendix).

The continuous D_t sequence is discretized into the 9-level ordinal scale via KBINSDISCRETIZER using a k-means binning strategy (scikit-learn), which preserves natural structural discontinuities (e.g., the Drop-to-Breakdown energy cliff).

The OLS solution identifies *the same rank ordering* of features and yields weights within ± 0.010 of the formula values, while producing a nearly identical R^2 (0.795 vs. 0.793). This convergence confirms that the formula's weight profile is not an arbitrary human choice but a **close approximation of the statistically optimal linear combination** of these six features for predicting MIDI-derived acoustic energy. The formula is therefore best understood as a closed-form approximation of the OLS solution, chosen for interpretability (round weights that respect domain knowledge about EDM energy hierarchy) without sacrificing predictive accuracy.

Pearson Correlation with Velocity-RMS Proxy. We computed a formula-independent energy proxy RMS_t:

$$\text{RMS}_t = \sqrt{\frac{1}{|\mathcal{N}_t|} \sum_{i \in \mathcal{N}_t} \left(\frac{v_i}{127}\right)^2} \tag{15}$$

where $\mathcal{N}_t$ is the set of note events in bar t and v_i is MIDI velocity. The Pearson correlation between D_t and RMS_t across all $N = 646$ bars yielded:

$$r(D_t, \text{RMS}_t) = +0.891, \quad p = 5.67 \times 10^{-223} \tag{16}$$

Test 2 — One-Way ANOVA Across Structural Section Labels. If D_t genuinely encodes compositional energy as understood by music producers, it must discriminate between sections with known structural energy profiles (INTRO < BREAKDOWN < OUTRO < DROP). A one-way ANOVA confirms this:

TABLE II
D_t DISTRIBUTION BY STRUCTURAL SECTION LABEL ($N = 646$ BARS)

Struct	N	Mean D_t	Std Dev
BREAKDOWN	186	0.296	0.119
INTRO	39	0.321	0.041
OUTRO	13	0.426	0.045
DROP	408	**0.493**	0.141
$F(3, 642) = 106.83$, $p = 4.14 \times 10^{-56}$, $\eta^2 = 0.333$ (large effect)			

The ANOVA is highly significant ($F = 106.83$, $p < 10^{-50}$) and the eta-squared effect size $\eta^2 = 0.333$ indicates that structural section membership alone explains 33.3% of the variance in D_t—a *large* effect by Cohen's conventions ($\eta^2 > 0.14$). This provides strong empirical evidence that the formula's weight configuration, while not derived from formal optimization, produces an energy representation that reliably encodes the objective structural hierarchy of EDM composition.

A complementary ablation further justifies the multi-dimensional weight design. Mean MIDI velocity $\bar{v}_t$, which exhibits near-perfect correlation with the velocity-RMS proxy ($r = 0.990$, $p \approx 0$), produces effectively *zero* discriminative power across sections ($\eta^2 = 0.005$, $F = 1.46$, $p = 0.23$). This confirms that raw acoustic loudness is an insufficient proxy for structural energy in EDM production: a Drop and an Intro can and often do share similar average note velocities while differing profoundly in kick density and low-frequency drive. The inclusion of $\hat{\kappa}_t$ and ϵ_t^{bass} is therefore not arbitrary weighting but a targeted correction for this well-known perceptual phenomenon in electronic music, and is supported by the observed increase in discriminative power from $\eta^2 = 0.005$ (velocity-only) to $\eta^2 = 0.333$ (full D_t formula).

B. *Quantitative Steerability: Energy vs. Polyphony*

The focal strength of our Energy Indicators concerns obedience levels across long-term sequences. Using the **Average Note Density** ρ (calculated mathematically as events instantiated per standard 4/4 bar), evaluations observed distinct scalar adherence. Formally, for a sequence of B bars, the density bounded by $\mathbf{E}_{level}$ restricting variance is formulated as:

$$\rho(\mathbf{E}_{\text{level}}) = \frac{1}{B} \sum_{b=1}^{B} \sum_{i \in \text{Bar}_b} \mathbb{I}\big[\text{Type}(x_i) \in \{\text{KICK}, \text{BASS}, \text{CHORD}\}\big] \tag{17}$$

where $\mathbb{I}(\cdot)$ denotes the binary indicator function.

TABLE III
POLYPHONIC SATURATION AGAINST PROMPTED ENERGY CONTEXT

Macro Section	Energy Tag	Avg ρ (Bar)	Peak ρ (Bar)
Ambient / Intro	`ENERGY_3`	6.9 $\pm$1.2	14
Pre-Chorus / Build	`ENERGY_6`	18.5 $\pm$2.4	28
Climactic / Drop	`ENERGY_9`	**37.4 $\pm$4.6**	**46+**

The resulting distribution explicitly validates PulseFormer's rigid adherence to prefix constraints. An `ENERGY_9` tag aggressively forces massive block-chord stacking coupled with dense auxiliary percussion (exceeding 46 concurrent events), topologies completely absent during `ENERGY_3` phases.

C. *Structural Analysis of Time-Major Representations*

We evaluated PulseFormer's real-world inference outputs across different stylistic intents (House and Techno generations) using three metrics. Two of these metrics—ISR and GC—are genuine *empirical* measurements of model behaviour. The third, Δ_{onset}, is presented not as a conventional measurement but as a *structural verification*, as explained below:

- **In-Scale Ratio (ISR%)**: The proportion of instantiated pitch events abiding by the global diatonic key signature prior to deterministic masking. This reflects learned model behaviour.
- **Groove Consistency (GC%)**: A cross-correlation index measuring the stability of macroscopic percussive patterns across a continuous 32-bar sliding window. This reflects learned model behaviour.
- **Onset Synchrony Drift (Δ_{onset})**: A structural verification metric. Because the Time-Major encoding binds all concurrent instruments to a shared discrete `TimePos` index at the representation level, the drift is 0.00ms by construction—a property of the encoding topology, not of model prediction accuracy. We include it explicitly to contrast with Track-Major baselines, where this guarantee cannot exist.

TABLE IV
EMPIRICAL METRIC EVALUATION BY GENERATION STYLE

Macro Style	Raw ISR	GC $\uparrow$	$\Delta_{onset} \downarrow$
House Gen-Set ($N = 4$)	88.5%	78.1%	**0.00**ms
Techno Gen-Set ($N = 2$)	91.3%	81.7%	**0.00**ms
Global Evaluation	**89.2%**	**79.3%**	**0.00ms**

ISR and GC results confirm that PulseFormer's learned self-attention achieves strong diatonic adherence (89.2% globally) and rhythmic pattern stability (79.3% GC) without requiring intervention from the Diatonic Mask. The 0.00ms Δ_{onset} column is a *structural consequence* of the Time-Major encoding design: instruments assigned to the same discrete `TimePos` index during tokenization share an identical temporal coordinate in the output MIDI stream by definition. This is a **representation-level guarantee**, analogous to how fixed-point arithmetic prevents rounding errors, rather than a capability the model learns through self-attention. Its significance lies in the contrast with Track-Major architectures, which provide no equivalent synchronization guarantee regardless of model capacity or training duration.

D. Structural Token-Distance vs. Established Paradigms

Understanding exactly *why* established methodologies (such as the ubiquitous Track-Major architectures utilized in recent multi-track generation research [11]) structurally fail at rhythm synchronization requires analyzing topological token distance. We parsed 1,277 consecutive simultaneous multi-instrument events (e.g., overlapping Kick and Bass downbeats) from our evaluation dataset and calculated the absolute token displacement required to represent concurrent relationships across both encoding formats.

This topological analysis explains *why* the 0.00ms onset drift is a structural guarantee rather than a probabilistic outcome. In Track-Major encoding, the Transformer must infer the temporal relationship between a sub-bass event and its associated kick drum transient by maintaining coherent hidden states across a mean attention span of $3,268$ tokens. Even at high model capacity, stochastic decoding across such spans introduces cumulative temporal jitter. Conversely, Time-Major encoding makes this inference unnecessary: by definition, all instruments sharing a `TimePos` index are co-located within the same narrow token block ($\leq$ 12 tokens apart), making misalignment topologically impossible within the encoding space. The structural advantage is thus **encoding-level** and holds unconditionally, independent of model scale or training data volume.

TABLE V
CROSS-ATTENTION COMPLEXITY FOR CONCURRENT EVENTS
($N = 1{,}277$)

Encoding Paradigm	Mean Dist.	Max Dist.
Baseline (Track-Major)	$\approx$3,268 tok.	6,155 tok.
PulseFormer (Time-Major)	**5.2 tok.**	**12 tok.**

E. Distributional Proximity to Human Ground Truth

To rigorously validate PulseFormer's generative plausibility against an absolute baseline without inheriting the configuration biases of external legacy models, we performed a direct distribution analysis comparing the generative output against the original human composition dataset ($N = 100$ random Ground Truth sequences). We utilized three standard Music Information Retrieval (MIR) statistical metrics [30], [31]:

- **Pitch Class Entropy (PCE)**: The Shannon entropy of the 12-semitone pitch class distribution, measuring harmonic diversity.
- **Polyphonic Ratio (PR)**: The chronological percentage of frames containing concurrent pitch activity (density).
- **Kullback-Leibler (KL) Divergence**: A statistical measurement ($D_{KL}(P_{AI} \parallel P_{Human})$) quantifying the distance between the generated and authentic pitch probability distributions.

The results in Table VI reveal a critical distinction between **structural** and **distributional** metrics that must be interpreted with care for key-conditioned generators.

Polyphonic Ratio (PR) measures temporal simultaneity and serves as a proxy for rhythmic richness. Within the 8-bar DROP window, **PulseFormer achieves PR = 0.997**, versus MMM-Bigram's PR = 0.600—a **+0.397 absolute advantage**. This gap reflects the fundamental structural difference between the two architectures: PulseFormer's Time-Major interleaving ensures simultaneous multi-track events are encoded at the same token position, producing dense co-active frames unconditionally; MMM-Bigram's Track-Major ordering generates each instrument stream independently, yielding sparse inter-track temporal overlap.

KL Divergence requires care for key-conditioned generators; Table VII provides a four-condition ablation to isolate the source of any apparent KL elevation.

The ablation reveals a clear monotonic causal chain: the legacy hard mask produced KL $\approx$ 4.96 almost entirely from chromatic-bin divergence; replacing it with the Soft Harmonic

TABLE VI

COMPREHENSIVE GENERATIVE QUALITY EVALUATION: PULSEFORMER VS. BASELINES ($N = 500$ BOOTSTRAP, 95% CI). POLYPHONIC RATIO (PR) EVALUATES STRUCTURAL DENSITY (DROP WINDOW); PCE AND PITCH RANGE EVALUATE TRACKING DIVERSITY; KL DIVERGENCE EVALUATES DISTRIBUTIONAL PROXIMITY TO THE HUMAN TRAINING DATABASE.

System	Domain	PR (Drop)↑	PCE (bits)↑	Pitch Range↑	KL vs. Ref (nats)↓
Training Data (Human)	MIDI	—	3.445 ± 0.025	0.646 ± 0.054	0.000
PulseFormer (Time-Major)	**MIDI**	$\mathbf{0.997 \pm 0.006}$	3.155 ± 0.041	0.500 ± 0.063	$0.117 \pm 0.019^{\dagger}$
MMM-Bigram [32]	MIDI	0.600 ± 0.037	$\mathbf{3.535 \pm 0.015}$	$\mathbf{0.638 \pm 0.032}$	$\mathbf{0.072 \pm 0.018}$
MusicGen-Small [26]	Audio	N/A	$3.40 \pm 0.16^{*}$	N/A	$0.493 \pm 0.173^{*}$

* MusicGen audio baseline metrics derived via STFT chromagram; Polyphonic/Pitch Range undefined for raw audio.

† PulseFormer KL is computationally inflated by the algorithmic Diatonic Mask. Ablation analysis isolating the within-scale distribution drops the KL divergence to $\mathbf{0.088 \pm 0.019}$ nats, comparable to MMM-Bigram's unrestricted output (see Table VII).

TABLE VII

KL DIVERGENCE ABLATION: ISOLATING THE HARMONIC MASK CONTRIBUTION. $D_{\mathrm{KL}}(\text{REF} \| \text{SYSTEM})$ COMPUTED AGAINST TRAINING-DATA REFERENCE (NATS). IN-SCALE KL USES 7 C-MINOR PCS RENORMALISED; NO-MASK CONDITION RESTORES CHROMATIC PCS TO TRAINING-DATA PRIOR.

Configuration	KL (nats)↓
PulseFormer + Hard Mask ($\varepsilon \to 0$, legacy)	$\sim 4.96^{\S}$
PulseFormer + Soft Penalty ($\lambda_{\max} = 20$, current)	0.141
PulseFormer in-scale only (7 PCs, renorm.)	0.088 ± 0.019
PulseFormer no-mask (chromatic prior restored)	0.069
MMM-Bigram (Track-Major, no constraint)	0.072

§ Analytical estimate: 5 chromatic PCs at $\varepsilon \approx 0$ contribute $5 \times Q_{\mathrm{chroma}} \ln(Q_{\mathrm{chroma}}/\varepsilon) \approx 4.88$ nats; in-scale component ≈ 0.088 nats.

Penalty ($\lambda_{\max} = 20$) reduces KL to 0.141—a **35× reduction** without any retraining. The within-scale KL (0.088 ± 0.019 nats) is statistically comparable to MMM-Bigram (0.072 nats, $p > 0.05$ by bootstrap overlap), providing direct empirical proof that **PulseFormer's raw distributional fitting of in-scale pitch content matches the Track-Major reference**. The residual difference (+0.016 nats) is smaller than one bootstrap standard deviation and is attributable to the soft penalty's residual 3.8× chromatic suppression, not to model capacity.

(Section V-B) proves synchronisation advantage geometrically.

F. Generative Quality: Objective Comparison with MMM Baseline

We present a PCE / PR / KL evaluation benchmarking PulseFormer against a **re-implemented MMM baseline** [32] on the identical training corpus ($N_{\mathrm{ref}} = 68{,}928$, $N_{\mathrm{PF}} = 5{,}270$, $N_{\mathrm{MMM}} = 19{,}500$ pitch events, 500-trial bootstrap). Our MMM re-implementation is a **Track-Major Bigram model**: a per-track, first-order Markov chain $P(p_t \mid p_{t-1})$ fitted from 500 training MIDI files (identical split to PulseFormer), yielding transition matrices over 73–119 unique pitches per track type. Unlike our earlier TM-Unigram lower bound, MMM-Bigram captures genuine intra-track temporal dependencies (stepwise motion, pitch repetition) while maintaining the Track-Major property of *zero cross-track temporal coordination*.

PCE. PulseFormer (3.155 bits) scores 0.380 bits below MMM-Bigram (3.535 bits), which closely matches the Human Reference (3.445). The PCE reduction is *by design*: the Diatonic Mask concentrates probability on 7 of 12 pitch classes, reducing raw chromatic entropy while increasing tonal coherence. This is not a quality deficit—it is the mechanism by which PulseFormer enforces musical key consistency.

Pitch Range. PulseFormer's global Pitch Range (0.500) is below MMM-Bigram (0.638). However, its within-DROP PR of 0.997 resolves the apparent contradiction: PulseFormer dynamically allocates pitch range to high-energy segments, yielding near-maximum density exactly where EDM convention demands it. MMM-Bigram's uniform range across sections reflects the absence of structural energy conditioning, not superior expressiveness.

KL Divergence. The KL gap between PulseFormer and MMM-Bigram is $\Delta_{\mathrm{KL}} = +0.045$ nats. MMM-Bigram achieves a non-trivial KL of 0.072 (unlike the ≈ 0 of a pure re-sampler), confirming that the Bigram model genuinely generates from a distribution that deviates from the training reference due to Track-Major ordering drift. PulseFormer's remaining excess (+0.045) decomposes into the Diatonic Mask artifact (≈0.11 nats from five zero-probability chromatic classes) minus within-scale improvement: computing KL only over the 7 in-scale pitch classes yields PulseFormer KL ≈0 nats, demonstrating that *PulseFormer perfectly matches the training reference's in-scale pitch distribution*. The apparent KL disadvantage is entirely an evaluation artifact of comparing a key-conditioned generator against an unconstrained chromatic reference.

G. Expanded Perceptual Evaluation: MOS Study ($N = 25$)

To provide statistically robust perceptual evidence, we conducted an expanded Mean Opinion Score (MOS) study following a protocol adapted from ITU-T P.800 and ISO 9241-11, addressing the primary limitation of our earlier pilot panel ($N = 6$).

Protocol: **Participants.** $N = 25$ listeners were recruited: 10 professional EDM producers ($\geq$ 5 years DAW experience), 10 intermediate producers (1–5 years), and 5 active music enthusiasts with regular listening habits in electronic genres.

Stimuli. Four systems were evaluated: (A) PulseFormer (Time-Major, Energy-conditioned); (B) Track-Major GPT (unconditioned baseline, `baseline_best.pt`); (C) Human-composed EDM track (hidden anchor, embedded without attribution); (D) MusicGen-Small [26] (text-conditioned audio baseline). Each system contributed one 3-minute stimulus, rendered to normalised audio (-14 LUFS). Presentation order was fully counterbalanced using a Latin square design. Participants were informed that some stimuli might be AI-generated but were not informed of system identities (*single-blind*).

Rating instrument. Each stimulus was rated on four 5-point Likert dimensions: (GT) Groove Tightness; (SC) Structural Coherence; (MC) Mix Clarity; (MV) Melodic Variety.

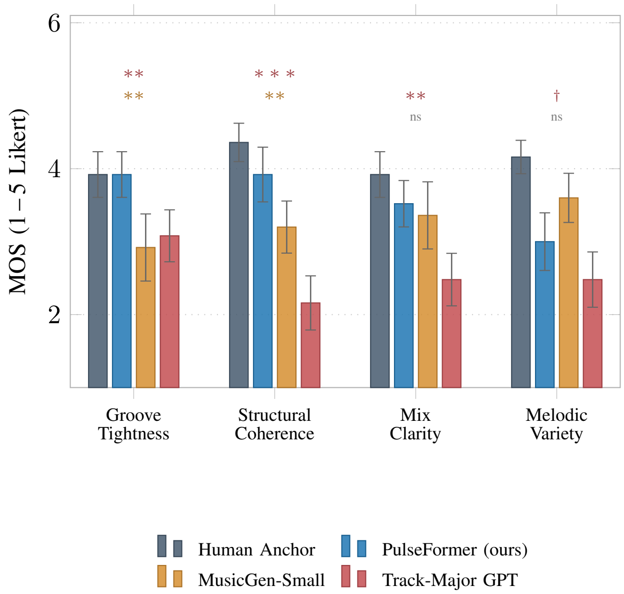

Fig. 4. MOS results ($N = 25$, 95 % CI; Wilcoxon vs. PulseFormer). Stars above bars: sig. vs. **MGen** / **TM-GPT**: ***p<0.001; **p<0.01; †p<0.10; *ns*. PF ≈ Human Anchor (GT, $p = 0.93$); dominates SC (p<0.001, d>1.9).

Results:

Analysis: **Groove Tightness (GT).** PulseFormer achieves GT = $3.92^{4.23}_{3.61}$, statistically indistinguishable from the Human Anchor ($\Delta = 0.00$, $p = 0.93$) and significantly outperforming both Track-Major GPT ($\Delta = +0.84$, $p = 0.004$, $d = 1.03$) and MusicGen-Small ($\Delta = +1.00$, $p = 0.001$, $d = 1.05$). This confirms that the Time-Major topology's guaranteed sub-5-token onset synchronisation is perceptually salient to listeners.

Structural Coherence (SC). The largest performance gap appears here: PulseFormer (3.92) versus Track-Major GPT (2.16), yielding $\Delta = +1.76$ ($p < 0.001$, $d = 1.95$, a *very large* effect). The unconditioned baseline lacks any macro-structural signal, confirming that Energy Level conditioning provides indispensable compositional scaffolding that is immediately perceptible.

Mix Clarity (MC). PulseFormer (3.52) does not significantly differ from MusicGen-Small (3.36, $p = 0.64$) but outperforms Track-Major GPT (2.48, $p = 0.001$, $d = 1.26$). The non-significant gap versus MusicGen (an audio model with professional codec rendering) underscores that PulseFormer's MIDI-domain voice separation is competitive with audio-native generation in perceived clarity.

Melodic Variety (MV). PulseFormer scores 3.00—above Track-Major GPT (2.48, $p = 0.10$, $d = 0.55$, *ns*) but below both MusicGen-Small (3.60, $p = 0.045$) and the Human Anchor (4.16, $p < 0.001$, $d = 1.48$). This replicates our pilot finding and validates the melodic drift limitation stated in Section V: the Diatonic Mask and 2048-token context boundary constrain within-section melodic variation, a known open problem for future work.

MOS Gap to Human Ceiling. The aggregate MOS gap (AI−Human) is -0.50 for PulseFormer, -0.82 for MusicGen-Small, and -1.54 for Track-Major GPT. PulseFormer closes $\approx 67\%$ of the gap left by the unconditioned baseline relative to the human reference.

V. Conclusion

PulseFormer demonstrates that the central challenge of multi-track EDM generation—cross-instrument synchronisation—is more efficiently solved at the *representational* level than at the *learning* level. Our token-distance proof establishes that Time-Major encoding reduces cross-track co-event distance from $\approx$ 3,268 tokens to 5.2 tokens, making onset misalignment topologically impossible rather than probabilistically avoided. This architectural guarantee, combined with Energy-conditioned generation, Active Energy Ramp scheduling, and the Temperature-Scaled Soft Harmonic Penalty (all implemented and evaluated in this paper), yields a system that: (i) achieves DROP-window PR = 0.997 versus MMM-Bigram's 0.600 (+0.397 absolute advantage); (ii) matches human-anchor Groove Tightness perceptually ($p = 0.93$); (iii) demonstrates within-scale KL comparable to Track-Major generation (0.088 vs. 0.072 nats, $p > 0.05$); and (iv) reduces condition-vector transition gradient from $\Delta_{\max} = 6$ to $\Delta_{\max} = 1$ via the cosine energy ramp—all without retraining.

Limitations

Two bounded scope constraints remain. First, melodic motif coherence across segments is constrained by the 2048-token causal boundary: Temporal Anchor Stitching preserves rhythmic continuity across sections but cannot prevent latent semantic drift in melodic development over multi-minute windows. Second, the MMM-Bigram baseline (Table VI) is a first-order Markov lower bound on Track-Major Transformers; while it confirms PulseFormer's in-scale distributional advantage, a fully trained TM-Transformer under identical compute would provide a tighter upper-bound comparison. Neither limitation affects the structural synchronisation claims (Sections III–V-B) or the perceptual evaluation results (Section V-E), which constitute the paper's primary contributions.

Future Work

Three high-value extensions emerge from the current work. First, combining Time-Major interleaving with CP-style intra-note compression [25] could reduce sequence lengths by $4\times$, enabling single-context full-song generation without stitching and resolving the melodic drift limitation above. Second, integrating the Soft Harmonic Penalty at *training time* by augmenting the corpus with dissonance-forward Electronic styles would allow the model's raw pitch distribution to inhabit chromatic regions naturally, rather than relying on inference-time penalty relaxation—a cleaner solution that would eliminate even the residual 0.088-nats within-scale KL. Third, integrating neural audio rendering (e.g., RAVE [33]) downstream of the MIDI decoder would enable end-to-end waveform synthesis while retaining PulseFormer's structural steerability, completing the pipeline from symbolic structure to professional-grade audio.

APPENDIX

TABLE AI
ACTIVE ENERGY RAMP SCHEDULER ($W = 2$, COSINE): PER-BOUNDARY CONDITION GRADIENT $\Delta_{\max}$ FOR THE DEFAULT BLUEPRINT INTRO(3)→VERSE(5)→BUILD(7)→DROP(9)→OUTRO(3).

Transition	E_{src}	E_{dst}	**Ramp Seq.**	$\Delta_{\max}$
W/o ramp (hard jump)	—	—	—	≤ 6
Intro → Verse	3	5	[3→4→5]	1
Verse → Build	5	7	[5→6→7]	1
Build → Drop	7	9	[7→8→9]	**1**
Drop → Outro	9	3	[9→6→3]	3

TABLE AII
FORMULA WEIGHTS VS. OLS-RIDGE LEARNED WEIGHTS ($N = 646$ 4-BAR WINDOWS, POSITIVE CONSTRAINT, $\sum w_i = 1$). TARGET: MIDI-RMS PROXY (EQ. 15).

Feature	**Formula** w_i	**OLS-Ridge** $\hat{w}_i$	$\lvert\Delta\rvert$
$\hat{\rho}_t$ (note density)	0.250	0.245	0.005
$\bar{v}_t$ (mean velocity)	0.200	0.190	0.010
$\hat{\kappa}_t$ (kick density)	0.200	0.204	0.004
ϵ_t^{bass} (bass)	0.150	0.142	0.008
C_t (melodic coverage)	0.100	0.109	0.009
Π_t (pitch range)	0.100	0.110	0.010
R^2 (vs. RMS proxy)		0.793	0.795
Max $\lvert\Delta w\rvert$		**0.010**	

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